Mandeep Khatri

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EDUCATION

Alabama A&M University, *Huntsville, AL* — ***Expected Graduation: May 2028***
Bachelor of Science in Computer Science — *GPA: 3.85/4.0*
Relevant Coursework: Data Structures & Algorithms, Programming I & II (Python), Computer Architecture, Software Engineering, Operating Systems, Linux with Application Programming, Distributed Systems

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, MATLAB, Go
Frameworks & Libraries: React, React Native, NestJS, Flask, pandas, NumPy, Polars, Streamlit, LangChain, FAISS
Cloud & Tools: AWS (EC2, Lambda), GCP (Vertex AI), Docker, PostgreSQL, Prisma, Git, GitLab CI, CUDA, Linux

WORK EXPERIENCE

Research Intern - International Space Weather Camp, *German Aerospace Center (DLR)* ***June 2026 – August 2026***
- Built a Python and Polars pipeline to clean and align **20M+ rows** of Wind and STEREO spacecraft data, yielding **752** quality-filtered CME events for **machine learning** model development.
- Engineered **56 statistical features** and evaluated Linear Regression, Random Forest, and Gradient Boosting across 20 seeds; the best model reached **0.93 Pearson correlation** for maximum magnetic field strength (Bt).

Teaching Assistant, *Alabama A&M University, Huntsville, AL* ***October 2025 – Present***
- Built a pytest autograder for CS104 (Python) and CS109 (C++), adopted by the TA team across **35-student lab sections**; cut per-assignment grading from **2.5h to 25min** and rewrote 4 lab handouts after student feedback.
- Started a weekly LeetCode workshop for freshmen prepping for internships; ran a **12-week DSA track** (arrays, hash maps, graphs, DP) on GitHub Classroom with mock-interview sessions, growing to **15+ regulars**.

Research Engineer - Autonomous System Lab, *Alabama A&M University, Huntsville, AL* ***January 2025 – Present***
- Led a rewrite of the drone perception pipeline from MATLAB to C++/CUDA; optimized parallel GPU processing and memory access, reducing per-frame latency from **84 ms to 27 ms** on Jetson Orin while sustaining **30 FPS**.
- Developed Python automation scripts & Gazebo regression-testing framework covering 14 scenarios; integrated tests into GitLab CI to automate routine perception testing & catch a frame-skip bug before flight testing.
- Proposed and built a Kalman-filter sensor-fusion module in C++ that smooths IMU + camera streams in the perception stack, cutting simulated trajectory jitter **38%**.

Software Engineering Intern, *Verisk, Remote* ***August 2023 – December 2023***
- Added new sensor types in the team's IoT telemetry service (Python, MQTT, AWS EC2); shipped a JavaScript filter on the Chart.js dashboard, a Flask CSV endpoint, **47 Pytest cases**, & helped Dockerize the service.
- Investigated slow REST API responses under JMeter load testing (**~300 RPS**) and worked with my mentor to identify a Postgres connection pool issue; after the fix, p95 API latency dropped from **~1.4s to ~820ms**.

PROJECTS

Internably - *Student Networking Platform* ***March 2026 – Present***
- Built a full-stack mobile application for college internship networking using **React Native, TypeScript, NestJS, PostgreSQL, and Prisma**; designed authentication with .edu verification, JWT refresh-token rotation, and real-time messaging through Socket.IO.
- Built **optimistic updates with TanStack Query** for likes and connection requests with rollback/cache synchronization; organized the app as an npm-workspaces monorepo with three shared packages.

DealScreen AI - *Industrial Real Estate Deal Screening Platform* ***Link Logistics Hackathon, August 2026***
- Built an end-to-end AI deal-screening pipeline using JavaScript, OpenAI APIs, webhooks, and n8n that parses commercial real-estate PDFs, extracts structured property and financial data, evaluates **5 investment-risk categories**, and sends results to an interactive dashboard.
- Added a **JavaScript validation layer** to check model output, handle missing financial inputs, and calculate vacancy, price/SF, and cap rate only when the required source data is available, keeping incomplete deal analyses from returning unsupported values.

Route Intelligence - *Food Rescue Platform* ***JPMorgan Chase Hackathon, April 2026***
- Processed **150K+ rescue records (6.67M lbs of food)** using Python and **pandas** across 4 cities; a ZIP-3 batching pass flagged **845 same-day pickup pairs**, unlocking **$106K in potential logistics savings**.
- Built a 4-tier risk score (recency, claim rate, pickup size) on a Folium + Streamlit map; **35% of rescues** fell into Watch/Elevated/High.

LEADERSHIP & ACTIVITIES

- **Resident Assistant** - *Alabama A&M University* — **June 2026 – Present**
- **Break Through Tech AI Fellow, Cornell Tech** - *Applied AI/ML program* — **May 2026 - Present**
- **NRF Foundation Ray Greenly Scholar** - *Competitive National Selection* — **January 2026**
- **Google Cloud Career Jumpstart** - *HBCU Fellow* — **May 2025 – August 2025**